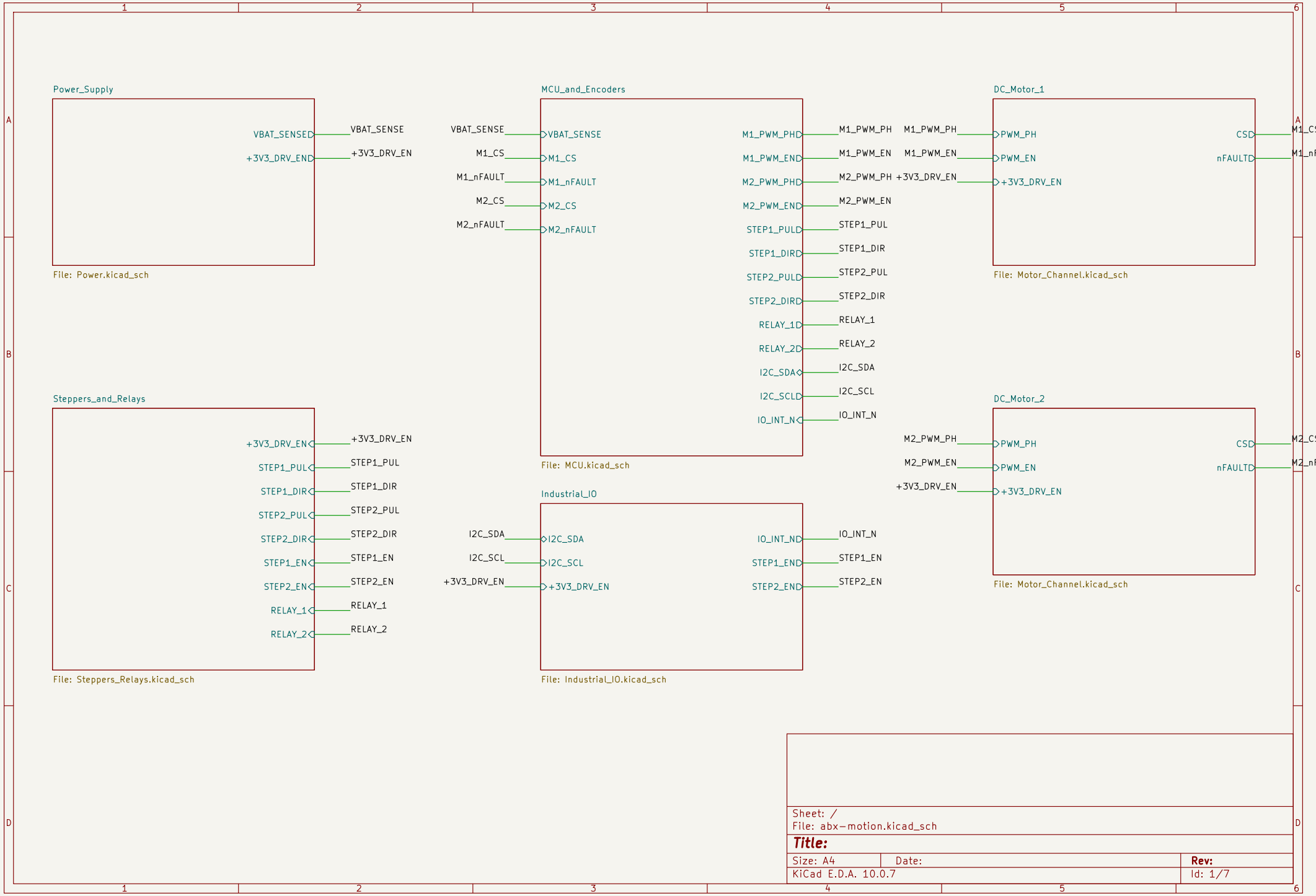
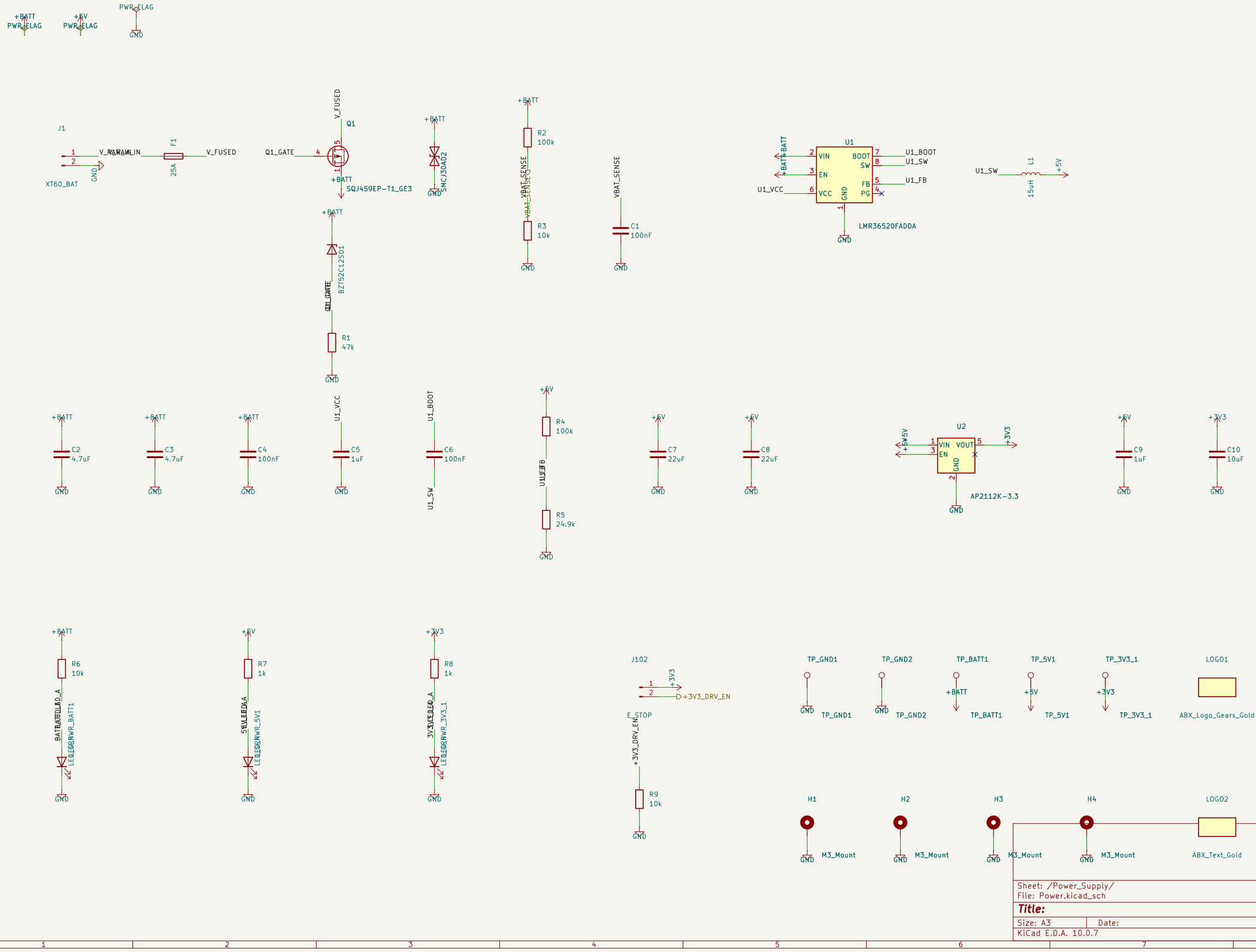
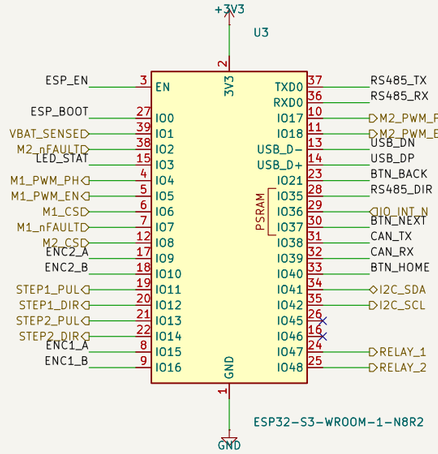

Power_Supply
VBAT_SENSE
+3V3_DRV_EN
VBAT_SENSE
+3V3_DRV_EN
File: Power.kicad_sch
MCU_and_Encoders
VBAT_SENSE
M1_CS
M1_nFAULT
M2_CS
M2_nFAULT
M1_PWM_PH
M1_PWM_EN
M2_PWM_PH
M2_PWM_EN
STEP1_PUL
STEP1_DIR
STEP2_PUL
STEP2_DIR
RELAY_1
RELAY_2
I2C_SDA
I2C_SCL
IO_INT_N
File: MCU.kicad_sch
DC_Motor_1
PWM_PH
PWM_EN
+3V3_DRV_EN
CSD
nFAULTD
M1_CS
M1_nFAULT
File: Motor_Channel.kicad_sch
Steppers_and_Relays
+3V3_DRV_EN
STEP1_PUL
STEP1_DIR
STEP2_PUL
STEP2_DIR
STEP1_EN
STEP2_EN
RELAY_1
RELAY_2
File: Steppers_Relays.kicad_sch
Industrial_IO
I2C_SDA
I2C_SCL
+3V3_DRV_EN
IO_INT_ND
STEP1_END
STEP2_END
File: Industrial_IO.kicad_sch
DC_Motor_2
M2_CS
M2_nFAULT
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Size: A4
Date:
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KiCad E.D.A. 10.0.7
Id: 1/7

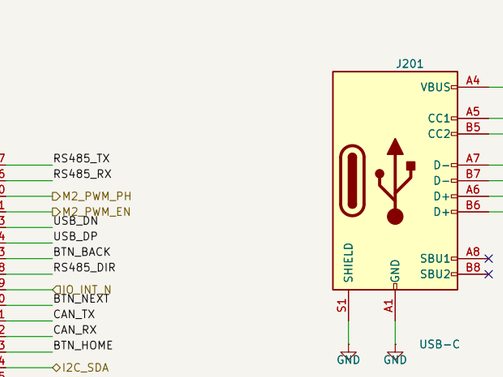
SHEET 1: BATTERY INPUT, REVERSE POLARITY PROTECTION, 5V BUCK & 3.3V LDO

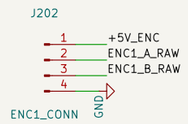


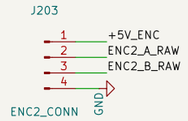
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Size: A3	Date:
KiCad E.D.A. 10.0.7	Id: 2/7
	Rev:

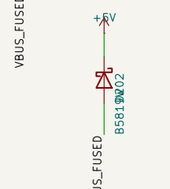
SHEET 2: ESP32-S3-WROOM-1, USB-C, 74LVC14A ENCODERS, CAN, RS-485, OLED & UI BUTTONS

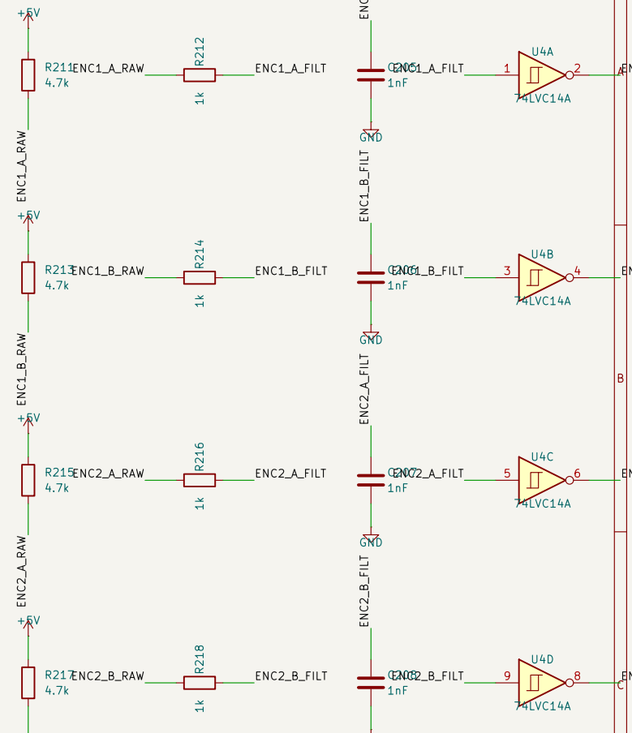


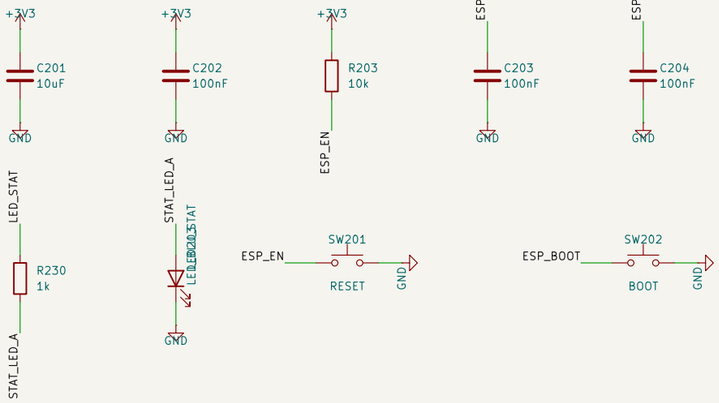


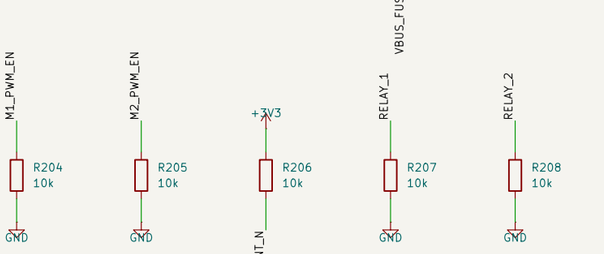


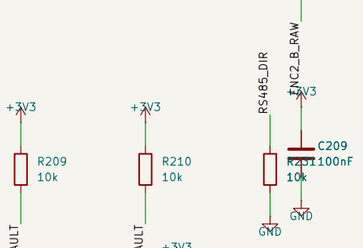


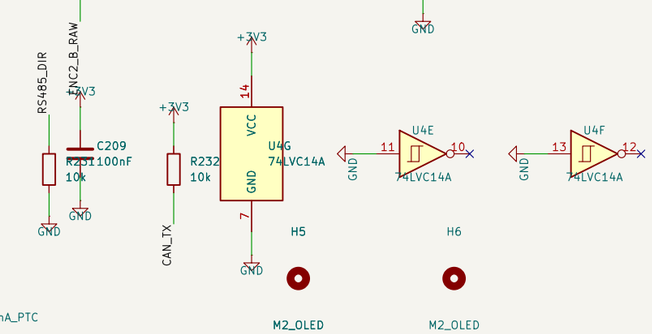


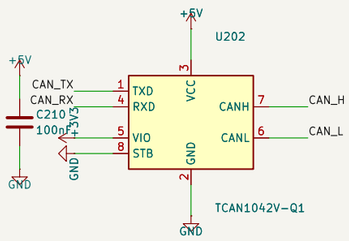


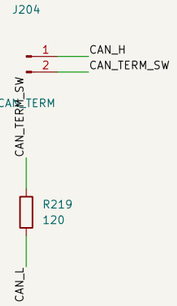


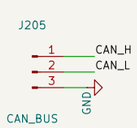




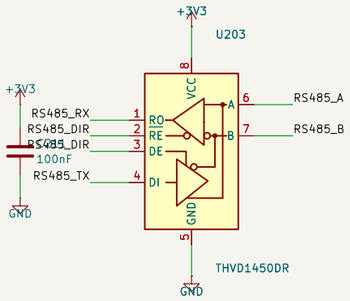


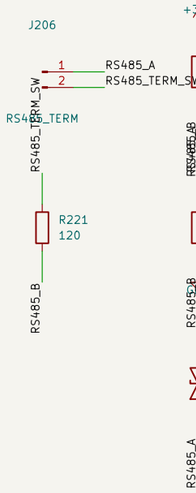


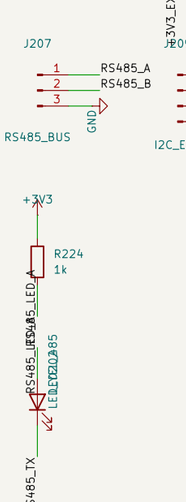


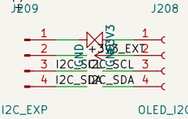


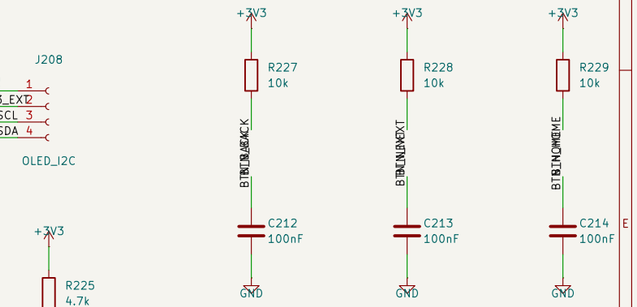


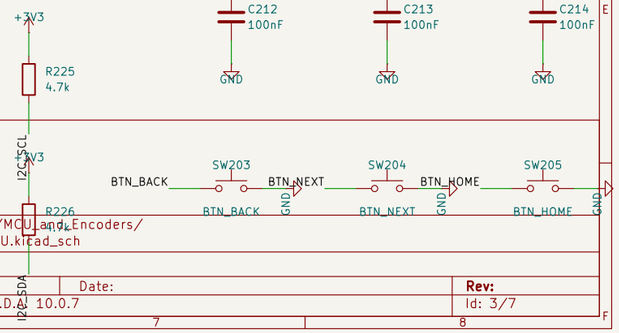










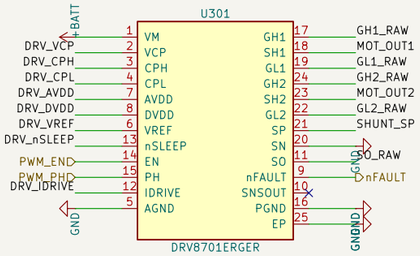


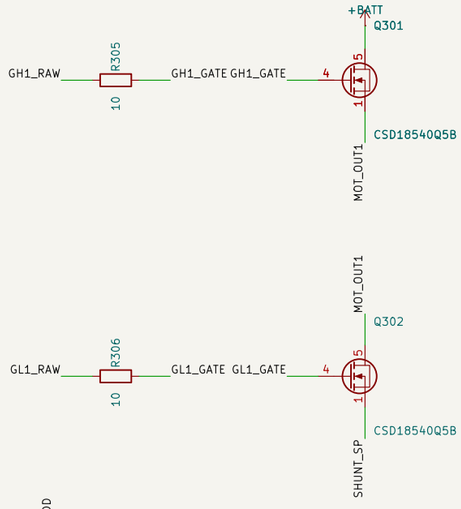
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File: MCU.kicad_sch

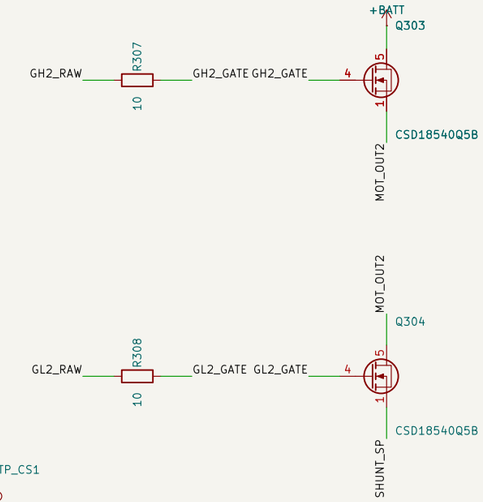
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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/7

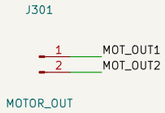
SHEET 3: 100W BRUSHED DC SERVO CHANNEL (DRV8701ERGER + 4x CSD18540Q5B 60V H-BRIDGE)

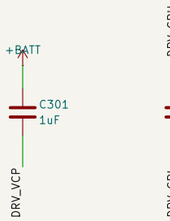




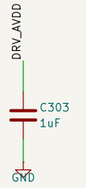


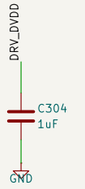




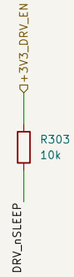


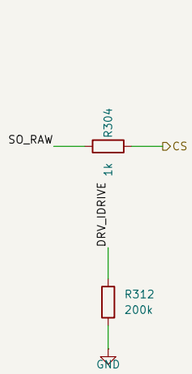




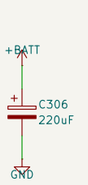


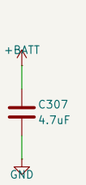


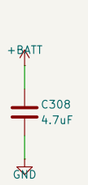


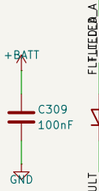








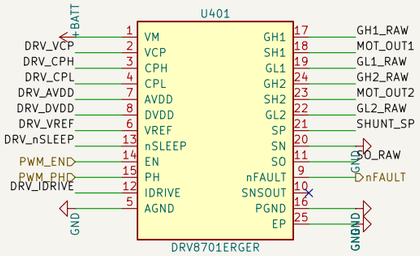


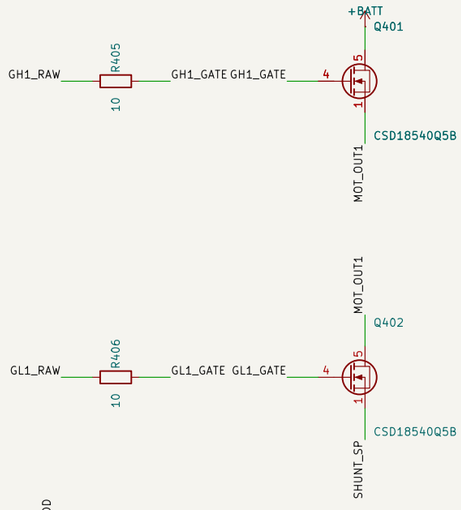


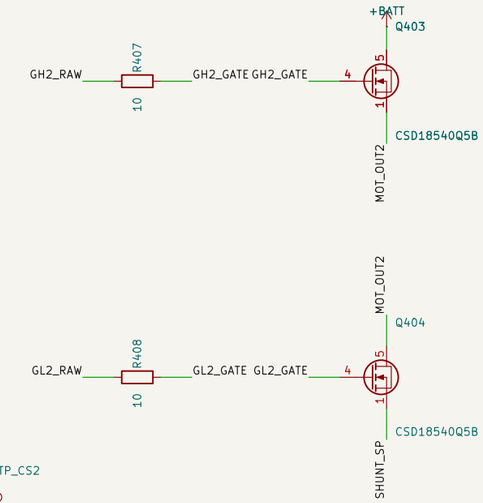


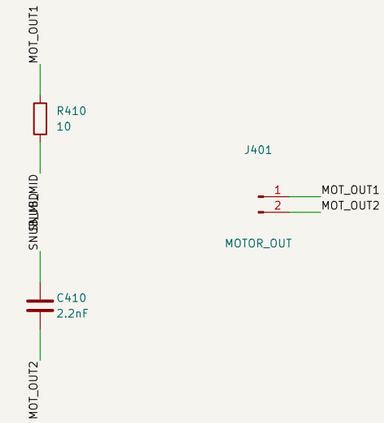
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Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 4/7

SHEET 3: 100W BRUSHED DC SERVO CHANNEL (DRV8701ERGER + 4x CSD18540Q5B 60V H-BRIDGE)



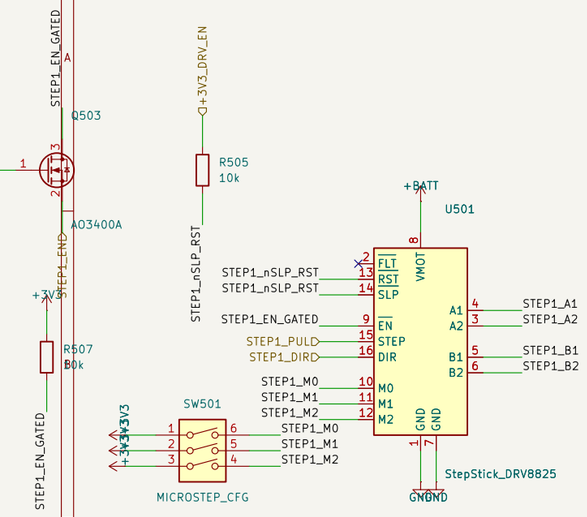


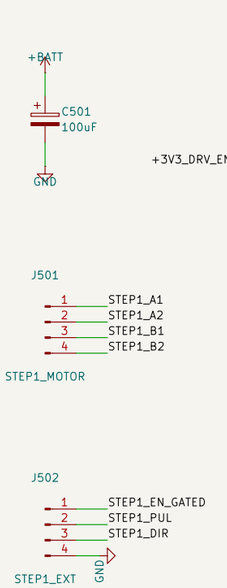


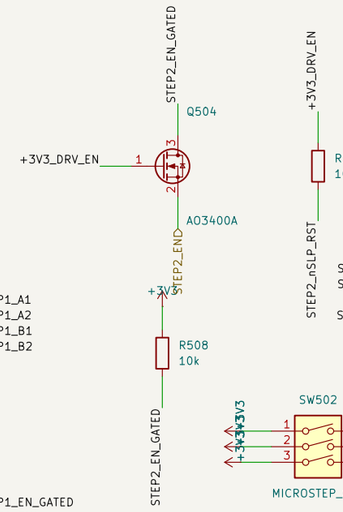


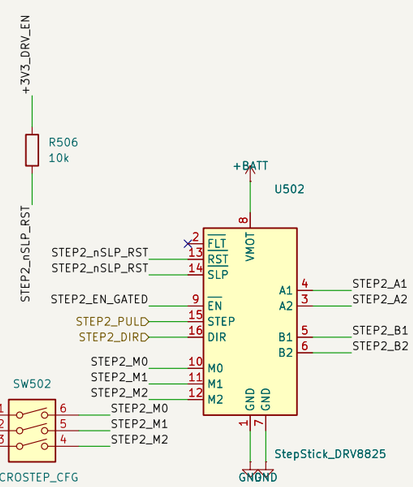
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KiCad E.D.A. 10.0.7		Id: 5/7

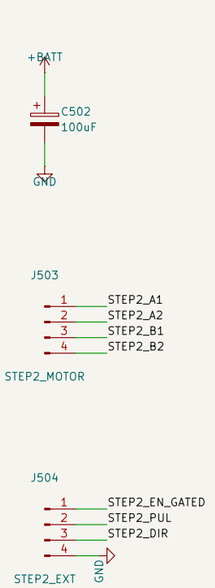
SHEET 4: DUAL STEPSTICK STEPPER CHANNELS & DUAL 5V OMRON G5LE RELAY OUTPUTS

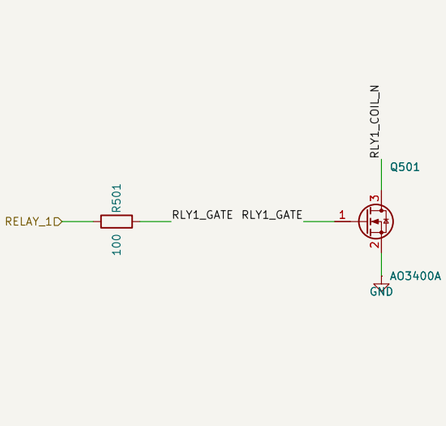




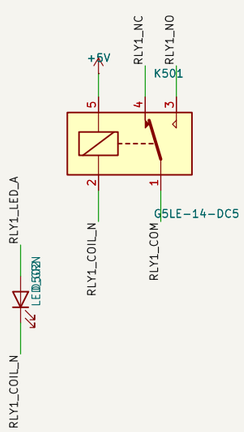


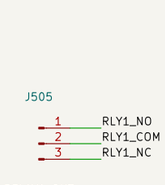


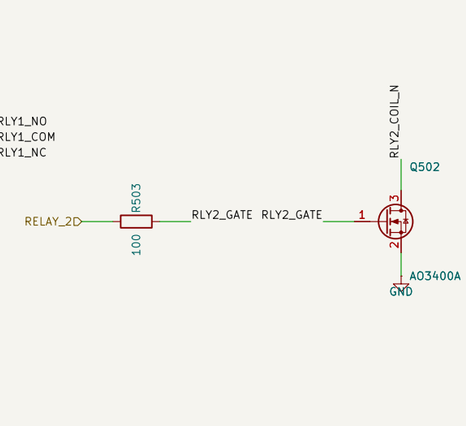




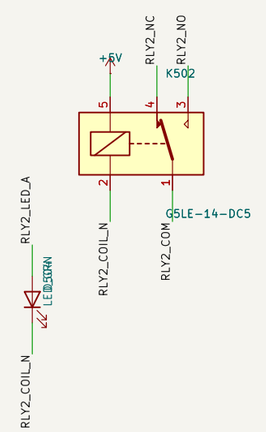


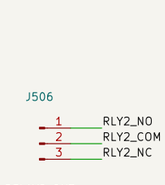












Sheet: /Steppers_and_Relays/
File: Steppers_Relays.kicad_sch

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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 6/7

SHEET 5: 6x 3-PIN JST-PH INDUSTRIAL I/O (8x LIMIT/GPIO + 2x 5V/500mA DC OUTPUTS) & PCA9555PW EXPANDER

J601
M1_MIN_RAW
M1_MAX_RAW
GND
B3B-PH-K-S

J602
M2_MIN_RAW
M2_MAX_RAW
GND
B3B-PH-K-S

J603
S1_MIN_RAW
S1_MAX_RAW
GND
B3B-PH-K-S

J604
S2_MIN_RAW
S2_MAX_RAW
GND
B3B-PH-K-S

+3V3
R601 4.7k
R602 4.7k
R603 4.7k
R604 4.7k
R605 4.7k
R606 4.7k
R607 4.7k
R608 4.7k
M1_MIN_RAW
M1_MAX_RAW
M2_MIN_RAW
M2_MAX_RAW
S1_MIN_RAW
S1_MAX_RAW
S2_MIN_RAW
S2_MAX_RAW

R609 1k
R610 1k
R611 1k
R612 1k
R613 1k
R614 1k
R615 1k
R616 1k
M1_MIN_RAW → M1_MIN_IN
M1_MAX_RAW → M1_MAX_IN
M2_MIN_RAW → M2_MIN_IN
M2_MAX_RAW → M2_MAX_IN
S1_MIN_RAW → S1_MIN_IN
S1_MAX_RAW → S1_MAX_IN
S2_MIN_RAW → S2_MIN_IN
S2_MAX_RAW → S2_MAX_IN

C602 10nF
C603 10nF
C604 10nF
C605 10nF
C606 10nF
C607 10nF
C608 10nF
C609 10nF
M1_MIN_IN
M1_MAX_IN
M2_MIN_IN
M2_MAX_IN
S1_MIN_IN
S1_MAX_IN
S2_MIN_IN
S2_MAX_IN
GND

U601
+3V3
I2C_SCLD
I2C_SDA
IO_INT_N
C601 100nF
GND

SCL
SDA
INT
VDD
VSS
A0
A1
A2
GND
IO0_0 M1_MIN_IN
IO0_1 M1_MAX_IN
IO0_2 M2_MIN_IN
IO0_3 M2_MAX_IN
IO0_4 S1_MIN_IN
IO0_5 S1_MAX_IN
IO0_6 S2_MIN_IN
IO0_7 S2_MAX_IN
IO1_0 OUT1_CMD
IO1_1 OUT2_CMD
IO1_2 STEP1_EN
IO1_3 STEP2_EN
IO1_4 +3V3_DRV_EN
IO1_5
IO1_6
IO1_7
PCA9555PW,118

+5V
F5 500mA_PTC
+5V_OUT

OUT1_CMD
R617 1k
GND

OUT1_CMD
R619 100
OUT1_GATE
Q601
OUT1_SW
AO3400A
GND

+5V_OUT
D601 B5819W
OUT1_SW

J605
+5V_OUT
GND
OUT1_SW
B3B-PH-K-S

OUT2_CMD
R618 1k
GND

OUT2_CMD
R620 100
OUT2_GATE
Q602
OUT2_SW
AO3400A
GND

+5V_OUT
D602 B5819W
OUT2_SW

J606
+5V_OUT
GND
OUT2_SW
B3B-PH-K-S

+3V3
R621 10k
STEP1_EN

+3V3
R622 10k
STEP2_EN

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